

Chaebin Kim

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EDUCATION

Virginia Tech

Blacksburg, VA

Bachelor of Science in Computer Science, Minor in Human-Computer Interaction and Mathematics

May 2026

- GPA: 3.65/4.0 | Dean's List: Fall 2023, Fall 2024, Spring 2025, Fall 2025, and Spring 2026
- Coursework: HCI, Intro to AI, Cloud Dev, GUI Programming, Data Structures and Algorithms, Mobile Dev

EXPERIENCE

Undergraduate Research Intern

July 2025 – Aug 2025

Queen Mary University of London - CIS Lab

London, United Kingdom

- Designed and built an end-to-end multimodal AI evaluation pipeline integrating LISA (segmentation), DAM (captioning), and a LLaMA-based critic to assess spatial accuracy (IoU) and semantic similarity – enabling systematic model error pattern analysis across diverse test conditions
- Resolved evaluation score distortion from scale mismatch between IoU and text similarity metrics by redesigning the normalization strategy, producing stable quantitative outputs
- Built an interactive Gradio interface to visualize segmentation outputs, compare predictions with targets, and generate prompt refinement feedback

Undergraduate Research Assistant

October 2025 – Present

Virginia Tech - DBWorkout

Blacksburg, VA

- Traced full data request flow across a 20+ person codebase to identify schema bottleneck, then redesigned PostgreSQL schema with a reusable Question Bank model and targeted indexing
- Rebuilt instructor filtering UI in React/TypeScript with automatic filter application and pagination; migrated API test suite from Postman to Bruno and validated CRUD operations and error handling across all endpoints

Undergraduate Teaching Assistant

Aug 2025 – May 2026

Virginia Tech - Intro to Software Design (CS1114)

Blacksburg, VA

- Supported 400+ students in weekly lab sessions and office hours, diagnosing Java logic errors and clarifying OOP concepts

Vice President & Web Master

Jan 2025 – June 2026

Korean Student Association at VT (VTKSA)

Blacksburg, VA

- Led operations and event coordination across a 7-person executive team as Vice President; managed club website and job board for Korean international students as Webmaster

PROJECTS

LLM-Guided Segmentation Assessment | *Python, Gradio*

- Built end-to-end evaluation pipeline measuring IoU-based spatial accuracy and LLaMA-based text similarity; restructured prompting strategy after initial outputs failed to reflect intended evaluation criteria, achieving consistent result convergence
- Designed scoring normalization logic to reconcile scale differences between spatial and semantic metrics

Simulacra – LLM Agent Social Simulation | *Python*

- Integrated LLM backend into an existing agent simulation system and built a longitudinal survey system (daily/weekly/monthly) to measure how governance structures affect behavior outcomes across 100 agent profiles

DBWorkout - SQL Learning Platform | *Python, React, PostgreSQL*

- Contributed schema redesign, filterable instructor views, and full API validation to a web-based SQL learning platform actively used in live university courses

Feather Finder - Birdwatching Web App | *Java, HTML/CSS, AWS EC2*

- Built backend schema and intermediate transformation logic to reconcile mismatched response formats between eBird and Google Maps APIs; deployed on AWS EC2 in a 6-person team

TECHNICAL SKILLS

Languages: Python, Java, C, Swift, JavaScript, HTML/CSS, SQL

Frameworks/Tools: React, Node.js, PostgreSQL, Git, Docker, AWS EC2, Bruno, Postman, Linux/Bash

Other: English (Fluent), Korean (Native)